

TRIDENT-Luna

Triple-Redundant Integrated Design for Extraterrestrial Needs — lunar surface configuration

Status: design concept (not flight hardware) · **Oxygen:** 3.6 kg/day (≈ 4.3 crew-equivalents) · **Power:** 2.3 kW average · **Mass:** 324 kg (excl. shared power plant)

1. What TRIDENT-Luna is

TRIDENT-Luna is the lunar configuration of the TRIDENT ISRU platform: a life-support-scale oxygen plant that prioritizes maintainability over production rate. Three parallel process strings sit on a shared thermal mass so one string can be isolated and cooled for servicing while the others keep producing (warm-swap). The lunar variant pivots the feedstock chain from the Martian atmosphere to **polar water ice**: ice-bearing regolith is mined and heated, the captured water is electrolyzed, and oxygen is delivered to life support while hydrogen is stored as a by-product.

2. What changes from Mars, and why

- **Feedstock:** no usable atmosphere on the Moon — water ice from permanently shadowed regions replaces atmospheric CO₂; steam electrolysis replaces SOEC CO₂ co-electrolysis.
- **Hydrogen:** the Mars core loop carries a hydrogen deficit supplied externally; the lunar loop produces 0.45 kg H₂/day as a by-product.
- **Methane:** carbon cannot be ISRU-closed on the Moon, so the baseline produces no methane (the mirror image of the Mars variant's open hydrogen item).
- **Power:** a 354 h equatorial night makes solar-battery storage non-viable at this load (~6,771 kg of Li-ion would be required; Mars needs only ~189 kg). Baseline: an allocation from a shared 40 kWe-class fission surface power plant.
- **Dust:** no gas phase for an electrostatic precipitator — the lunar variant uses electrodynamic dust shields (>90% removal in vacuum simulant tests) plus mechanical exclusion and seals.
- **Regolith oxygen** (~40–45 wt% of regolith) exists everywhere but costs ~20–25 kWh/kg O₂ to extract — carried as a contingency kit, not the baseline.

3. Interface and architecture

Stage	Function	Notes
Ice feedstock kit	mine and heat ice-bearing regolith; capture water vapor	~81 kg regolith/day at 5 wt% ice
Electrolysis strings ×3	2 H ₂ O → 2 H ₂ + O ₂ (SOEC-class, steam mode)	shared thermal mass; warm-swap
Storage	gaseous O ₂ buffer (3 days) + H ₂ by-product	liquefaction optional, not modeled
Dust management	electrodynamic shields + seals	no consumables
Power	~3.4 kWe allocation from shared fission plant	battery for transients only

4. Budgets (generated by tools/model.py)

Flows

Flow	Value
O ₂ delivered	3.6 kg/day
Water consumed (stoichiometric)	4.05 kg/day
Regolith processed	81.0 kg/day at 5 wt% ice
H ₂ by-product (stored)	0.45 kg/day

Energy

Stage	kWh/day
Ice mining + water capture	12.1
Electrolysis	23.6
Balance of plant	19.4
Total	55.1 kWh/day → 2.3 kW average

Mass

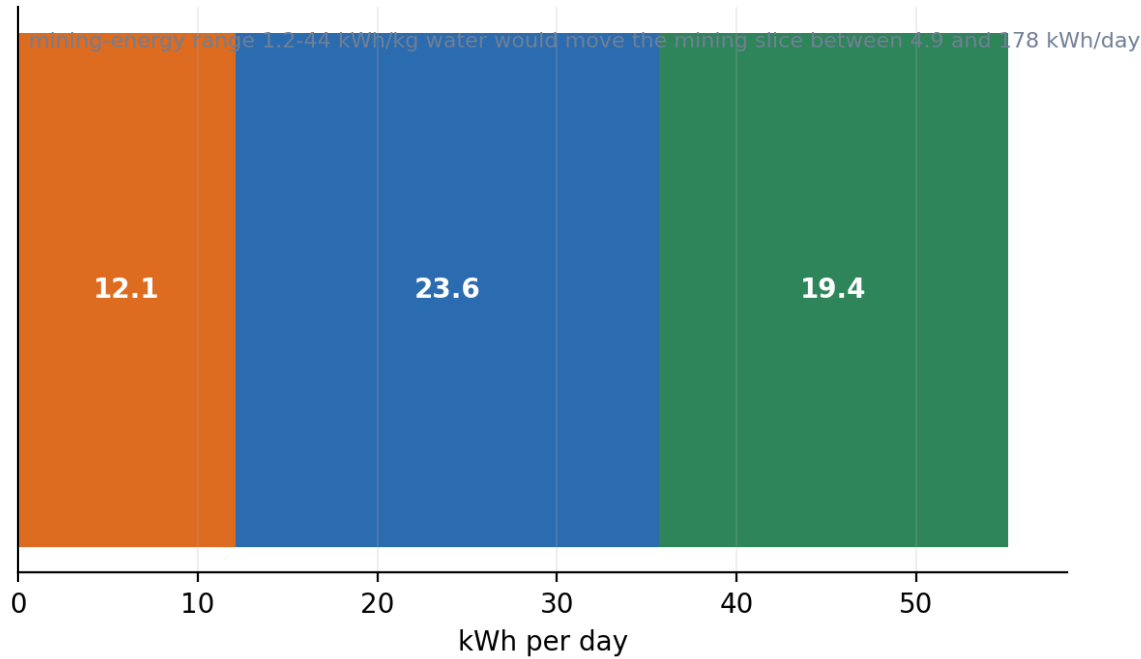
Item	kg
Processing core (strings, thermal, dust, structure, redundancy)	162

Ice feedstock kit	95
Gaseous O ₂ storage	43
Avionics + harness	24
Total	324 kg
Optional regolith-O ₂ contingency kit	+140 kg

Dominant uncertainty — ice-mining energy. Published estimates span 1.2 kWh/kg water (modeled thermal mining) to ~44 kWh/kg (lab-demonstrated, small scale). Across that band the plant's total energy moves between 47.8 and 221.1 kWh/day. The design target (55.1 kWh/day at 3 kWh/kg) assumes heat recuperation at scale and must be demonstrated.

5. Figures

TRIDENT-Luna daily energy: 55.1 kWh/day → 2.3 kW average



Concept estimates from tools/model.py (2026-10-01 run); expect $\pm 30\text{-}50\%$. Sources: variants/moon/SOURCES.md.

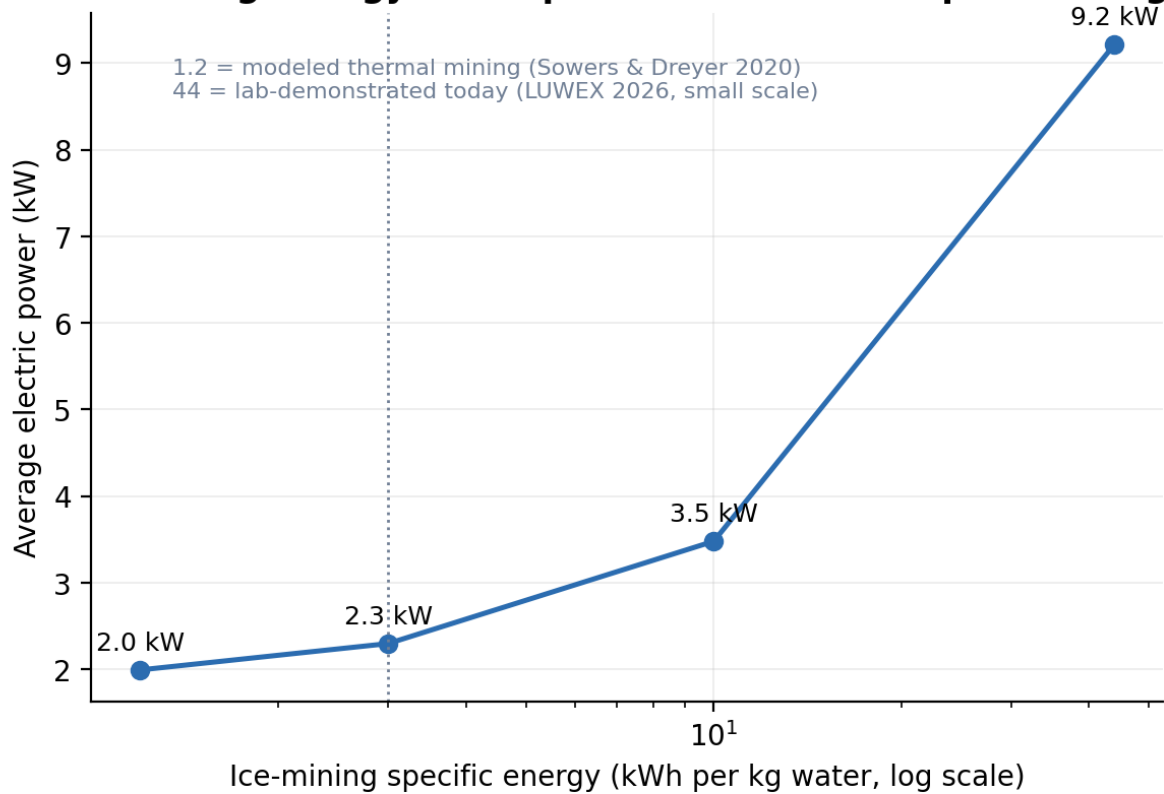
Ice mining
+ water capture

Electrolysis
(steam SOEC-class)

Balance of plant
(thermal, dust, avionics...)

Figure 1 — Daily energy by stage.

The mining-energy assumption dominates the power budget



Concept estimates from tools/model.py (2026-10-01 run); expect $\pm 30\text{-}50\%$. Sources: variants/moon/SOURCES.md.

Figure 2 — Plant power vs ice-mining specific energy.

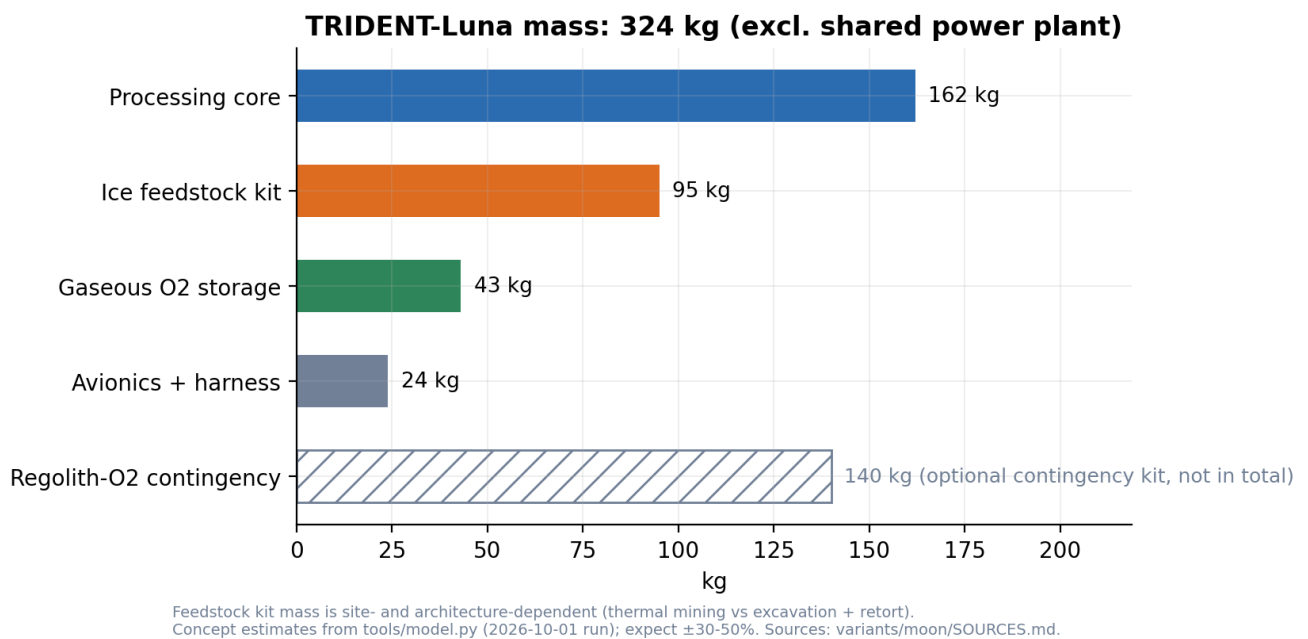
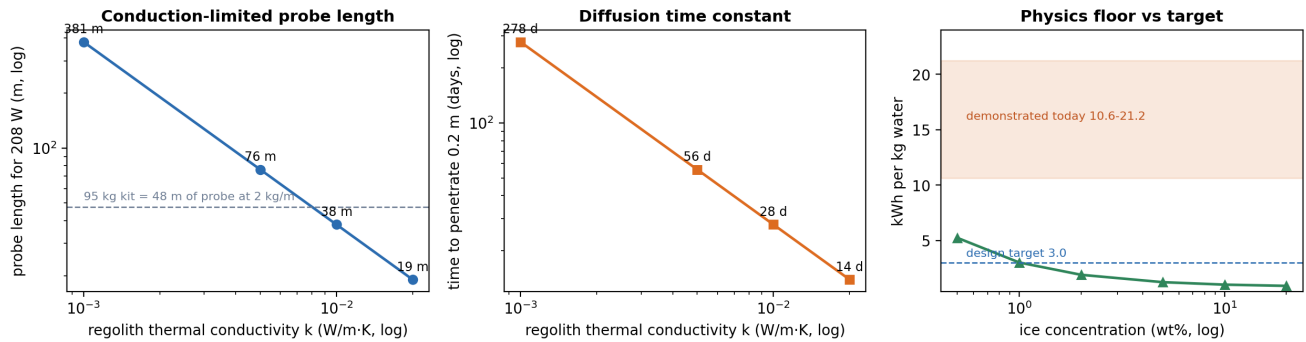


Figure 3 — Concept mass breakdown.

TRIDENT-Luna miner: conduction and energy screens (first-order)



Screen only, single-geometry proxies; heat demand = latent (133 W) + sensible (~75 W) at the design point. k range: Apollo heat-flow (0.001-0.025 W/m·K). Sources: variants/moon/miner-heat-transfer.md

Figure 4 — Miner heat-transfer screen: probe length, diffusion time constant, and the extraction-energy physics floor.

6. Open items

Screened in this revision: heat delivery (**miner-heat-transfer.md**, with the bench-test gate), operations (**ops-screens.md**), availability and spares (**availability-screen.md**), stowage break-even (**stowage-break-even.md**), and technology readiness (**trl-table.md**). The N-1 capacity margin is decided ($\geq 1.4\times$ per string). Still open: the physical bench tests, FMEA proper, condensation and rejection detail, and batching transients. Full ledger: **ASSUMPTIONS.md**.

7. Sources

All external figures: **variants/moon/SOURCES.md** (verified 2026-10-01) — NSSDC Moon Fact Sheet; Colaprete et al. 2010 (LCROSS); Mazarico 2011 / Gläser 2014 (illumination); Paige/Hayne 2010 (PSR temperatures); NASA/ESA (regolith oxygen, EDS, MRE); DOE 2024 (electrolysis); Sowers & Dreyer 2020 and LUWEX 2026 (ice mining); NASA NTRS 20220004670 (fission surface power).

8. Disclaimer

TRIDENT-Luna is a conceptual design for research, discussion and engineering evaluation only. It is not flight hardware, has not been built or tested, and must not be treated as a construction or operational plan. All performance figures are engineering estimates; real implementation requires independent detailed design, hazard analysis, qualification testing and professional engineering oversight.

Author: Nicholas Dean Perry · design concept, October 2026 · Generated from tools/model.py (run 2026-10-01). Verified by 11 regression tests and CI (github.com/luckyseoul/trident-platform).